



Drone Building Activity

Activity: Build a simple drone and test its manoeuvrability based on your understanding of drone forces and motion.

Materials Required:

- Foam board
- Four brushless motors
- Four propellers
- Flight controller
- Battery
- ESCs (Electronic Speed Controllers)
- Wires
- Tools (cutter, glue, soldering iron)

Instructions:

1. Design the Frame:

- Draw a simple cross or X shape on the foam board to serve as the drone's frame.
- Cut out the shape using a cutter.
- Mark the locations for the motors on the arms of the cross.

2. Mount the Motors:

- Drill holes in the foam board at the marked locations.
- Secure the motors to the frame using screws or glue.

3. Connect the ESCs:

- Connect each ESC to a motor using wires. Ensure correct polarity.
- Connect the ESCs to the flight controller according to the manufacturer's instructions.



4. Attach the Propellers:

- Attach the propellers to the motors, ensuring correct direction for lift.

5. Connect the Battery:

- Connect the battery to the flight controller's power input.

6. Calibrate the Flight Controller:

- Follow the flight controller's manual to calibrate the gyroscope and accelerometer.

7. Test the Drone:

- Place the drone on a flat surface.
- Turn on the power.
- Gradually increase the throttle to lift the drone.
- Practice controlling the drone's movements using the flight controller.

Tips:

- Start with smaller propellers for easier control.
- Use a battery with sufficient capacity to ensure longer flight times.
- Experiment with different flight modes and settings on the flight controller.
- Always fly in a safe and open area.